Neel Shah

Sports Pricing · Prediction Markets · Forecasting & Calibration

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SUMMARY

I build calibrated pricing models and the tools that check them. My project, CourtVision, prices **four sports (NBA, MLB, soccer, tennis)** and scores every model **leak-free and walk-forward against the devigged closing line**. The honest result: on moneylines it lands near the close but never beats it, and on totals and tennis it trails. I also built the **validation harness** (walk-forward with purge and embargo, game-clustered Diebold-Mariano tests, a two-corpus acceptance rule) that **caught and retracted six of my own inflated results**, and I publish the rejects. Execution is **paper only, behind six pre-registered go-live gates**; no real order has been placed. Earlier work on the same project covers broadcast-video player tracking.

SELECTED PROJECT: COURTVISION

Calibrated multi-sport forecasting engine, with the validation harness and paper-only execution controls as first-class components.
Evidence: github.com/neeljshah/court-vision/blob/master/docs/QUANT_BRIEF.md · github.com/neeljshah/court-vision/blob/master/EVIDENCE.md

Pricing quality, scored against the market

- Convert moneyline and over/under odds to fair probabilities with four **de-vig methods** implemented from scratch (including Shin, 1992), then score the forecasts by Brier against the devigged **sports-betting closing line** and against live Kalshi **prediction-market prices**.
- **Built paper-only risk controls:** an order lifecycle whose live path is unreachable behind three independent refusals, a fail-closed venue fee model, and a closing-line-value ledger with an off-market-price filter; no real order has been placed.
- Run a **forward paper desk** on Kalshi in-game markets (MLB, NBA, soccer, tennis, pro football, NCAA football) since 2026-10-06: a prospective schedule on 10 of 13 days, 113 games served including a 26-game slate on 2026-10-10, and a night receipt on 8 of those days.
- **Pregame vs the devigged close:** NBA moneyline Brier **0.1735 vs 0.1666** for the close (n=372 held out, 95% CI of the gap includes 0); MLB moneyline 0.2429 vs 0.2390 (n=13,992); soccer O/U 2.5 0.2465 vs 0.2390 (n=7,558); totals and ATP tennis trail by more. **Nothing beats the close.**
- **In-game:** conditioning on realized game state is sharper than a static pregame prior (**NBA Brier 0.209 to 0.159, MLB 0.241 to 0.126**, real-corpus out-of-sample). Where a live market exists (NBA end of Q1, Kalshi MLB and soccer ticks), the market is sharper than the model, and the verdicts are published.
- **Signals:** 0 of 60 candidate signal classes shipped; 513 verdicts were REJECT or DEFER. Candidate-specific gates included clustered DM, confidence intervals, null-shuffle tests, and Benjamini-Hochberg control.

Validation, and what did not survive it

- **Harness:** expanding-window walk-forward with a per-fold date assertion, **truncation-invariance** leak tests, acceptance only when a change beats baseline on **two independent corpora**, and a published minimum detectable effect for every market-relative row.
- **Published results:** pregame probabilities match or trail the devigged close and the in-game model is behind the live market, with n and 95% intervals; six withdrawn figures published with reasons at /evidence/retractions/; 16 findings on one template with limitations and verdict history.
- **Execution controls, paper only:** an order lifecycle whose live path is unreachable behind three independent refusals; a venue fee model that fails closed; a **CLV ledger with an off-market-price filter**, with its writer tested race-safe across two OS processes. The forward paper series began 2026-10-06 (0 of 8 required weeks).

Engineering & Data

- **Modeling and data:** per-stat **XGBoost** with **isotonic** calibration; four **de-vig methods** from scratch incl. **Shin (1992)**; a 151-file intelligence layer incl. a 291,625-pair player-vs-player matchup matrix; AI agents propose signals and the gate rejects them.
- **Private engine:** FastAPI with ~99 endpoints across 12 routers, 9 long-running daemons under a watchdog, a 430-module codebase, ~7,400-test pytest suite, CI/CD, 5 Docker images; an MCP server (interface documented publicly) answers queries from AI agents with cited artifacts.
- **Computer vision:** broadcast video to court coordinates on one consumer GPU: YOLOv8 detection, SIFT homography, Kalman+Hungarian tracking, OSNet re-ID reimplemented in PyTorch; output quality measured and documented, and tracking paused because its features carry zero importance in the production models.
- **Evidence site:** one unified, CI-gated site at https://neeljshah.github.io/court-vision/, built by one person directing AI coding agents; performance-budget and accessibility (axe) gates are hard CI steps with zero serious or critical violations; a PAPER ONLY desk record with receipts.

PROFESSIONAL EXPERIENCE

Data Analyst Intern, Sunsolor · *Tempe, AZ · 2025*
Built live Tableau / Power BI suites; cut manual reporting latency ~30%; automated daily market-trend pipelines in Python; data-quality audits across 5+ datasets; optimized SQL on 1M+-row datasets.

Data Annotation Specialist, DataAnnotation.Tech · *Remote · 2024–2025*
Labeled 10,000+ structured samples for production ML training & LLM fine-tuning; authored corner-case documentation cutting pipeline error variance ~20%; refined labeling schemas with data scientists.

Business Intelligence Analyst, Fortrex Securities · *Madison, WI · 2023–2024*
Built secure executive reporting interfaces (99.9% uptime); refined backend payment APIs & reporting pipelines for high-throughput billing; engineered real-time anomaly-detection alerting.

EDUCATION & CERTIFICATIONS

University of Iowa: three years of data-science coursework. Coursework: math stats, probability theory, machine learning, financial analytics.
Certifications: Certified Data Scientist (USDSI) · ML Specialist & Data Viz (Kaggle) · CBIAP (IABAC).

SKILLS

Methods: walk-forward / leak-free validation, calibration (isotonic), de-vigging (Shin), fractional-Kelly, Monte-Carlo simulation, shadow-logging, multi-agent orchestration
Languages: Python (expert), SQL, JavaScript/TypeScript (working)
ML/CV: PyTorch, OpenCV, YOLOv8/Ultralytics, ONNX, TensorRT, NumPy, Pandas, scikit-learn, XGBoost, LightGBM, statsmodels
Backend/Infra: FastAPI, WebSocket/SSE, PostgreSQL, SQLite, Docker, GitHub Actions, GCP, Tableau, Power BI, dashboards